Thursday. Aug 27

- Before moving onto CA (next section), there are a few other things that will be helpful for us to work with $\mathbb{R}$.

Defn. $\forall\, x \in \mathbb{R}$. define its **absolute value** $|x|$ as

$$|x| := \begin{cases} x, & \text{if } x \geq 0 \\ -x, & \text{if } x < 0. \end{cases}$$

Prop 3. Let $x, y \in \mathbb{R}$. and $a \geq 0$. then

(A). $|x| \geq 0$ (B). $|x| \leq a$ iff $-a \leq x \leq a$.

(C). $|xy| = |x| \cdot |y|$ (D). $|x+y| \leq |x| + |y|$ (Δ-ineq)

Pf. (A). (B) immediate by defn.

(C). Case-by-case analysis: $++, +-, -+, --$.

(D). Apply part (B). □

Prop 4. Let $x, y \in \mathbb{R}$. Such that $x \leq y + \varepsilon$, $\forall\, \varepsilon > 0$. Then $x \leq y$.

Pf. (This result is very useful when one wants to create some "ε room" in an argument. The Constructive type argument is also very commonly used.)

Suppose the claim is false. i.e. $x > y$.

Let $\varepsilon := \frac{x-y}{2} \in \mathbb{R}$. then one has $\varepsilon > 0$.

(By Axiom (O3). $x > y \Rightarrow x - y > y - y = 0$).

Observe that $y + \varepsilon = y + \frac{x-y}{2} = \frac{x+y}{2}$ (D). (A2). (A3)

We claim that $\frac{x+y}{2} < x$. (Indeed. $x = \frac{x}{2} + \frac{x}{2}$. apply (O3)).

Hence. $y + \varepsilon < x$. Contradiction. □

§1.2 The Completeness Axiom ("least upper bound" property)

Recall: one major drawback of $\mathbb{Q}$ is that it's not complete under the limit operation.

The goal of CA is to ensure that this won't happen in $\mathbb{R}$.

Defn. Let S be a subset of $\mathbb{R}$. ("$S \subset \mathbb{R}$"). We say S is **bounded from above** if there exists $m \in \mathbb{R}$ s.t. $s \leq m$, $\forall s \in S$. Such m is called an **upper bound** of S.

One can similarly define "**bounded from below**". "**lower bound**"

We say S is **bounded** if it is bounded from BOTH above and below.

Ex1. ① $S = \{1, 3, 5, 7, \cdots, 2k+1, \cdots\}$. bdd from below. not above.

Many different lower bounds: -100, 0, 1, $\cdots$

② $S = [0, 1)$. bdd from above and below.

Defn. If an upper (resp. lower) bound m of S is an element of S (i.e. $m \in S$), then m is called the **maximum** (resp. **minimum**) of S, denoted by $\max S$ (resp. $\min S$).

Rmk. Even for bdd set, max/min may not exist. For example. $S = [0, 1)$. $\max S$ doesn't exist. ($\forall x \in [0,1)$, $x < \frac{x+1}{2} < 1$. so x is not the max). In order to find a substitute for max/min that always exists for bdd set. we introduce the following concepts:

Defn. ① Let $S \subset \mathbb{R}$. bdd from above. Then $m \in \mathbb{R}$ is called the **least upper bound**, or "**supremum**", of S if:

(i) M is an upper bound of S (i.e. $M \geq s, \forall s \in S$)

(ii) $\forall M' \in \mathbb{R}$. if $M' < M$. then M' is not an upper bound of S (i.e. $\exists s \in S$ s.t. $M' < s$)

Such M. if exists. is denoted by $\sup S$.

② One can similarly define the greatest lower bound, or "infimum", of S. denoted by $\inf S$.

Rmk. (1) $\sup S$. $\inf S$ must be unique. (HW)

(2). $\sup S$. $\inf S$ may not be elements of S.

(3). If $\max S$ (or $\min S$) exists. then $\max S = \sup S$ ($\min S = \inf S$).

Ex. ① $S = \mathbb{N} = \{1, 2, 3, \cdots\}$. Unbdd from above. no Sup.

$\min S = \inf S = 1$.

② $S = \{x \in \mathbb{Q} : 0 \leq x < \sqrt{2}\}$. $\min S = \inf S = 0$. $\max S$ DNE. $\sup S = \sqrt{2}$. why?

③ $S = \phi$. bdd from above and below. all real numbers are its upper/lower bound.

min. max. inf. Sup DNE.

Thm. $\forall x, y \in \mathbb{R}$. $x < y$. $\exists r \in \mathbb{Q}$ s.t. $x < r < y$. (To be proved later)
(We also need to define $\sqrt{2}$ first)

Completeness Axiom (CA)

If $S \subset \mathbb{R}$. $S \neq \phi$. and S is bounded from above.

Then $\sup S$ exists.

Cor. If $S \subset \mathbb{R}$. $S \neq \phi$. and S is bounded from below.

Then $\inf S$ exists.

(In fact. these two statements are equivalent)

Pf. (Sketch). Define another set $W = \{-x : x \in S\}$. Show that $-(\sup W) = \inf S$. □

Rmk The CA doesn't hold in $\mathbb{Q}$.

For example, define $S = \{x \in \mathbb{Q} : 1 \le x \le \sqrt{2}\}$.

(We'll soon prove this is a real number.)

(Equivalently, $S = \{x \in \mathbb{Q} : 1 \le x^2 \le 2,\ x > 0\}$)

Our common sense tells us that one can find a sequence of rationals approaching $\sqrt{2}$ from below. Hence, $\sup S = \sqrt{2} \notin \mathbb{Q}$. (To justify: use "$\mathbb{Q}$ is dense in $\mathbb{R}$".)

* **Implications of Completeness Axiom.**

① $\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$ are both dense in $\mathbb{R}$.
($\mathbb{R} \setminus \mathbb{Q}$: "irrational numbers".)

② $x^2 = 2$ is solvable in $\mathbb{R}$. (i.e. $\sqrt{2}$ is a real number)

③ If a sequence of real numbers converges (in the "Cauchy" sense. will define in Chapter III), then it converges to a real number (This statement is called the "Cauchy convergence criterion", and is equivalent to CA.)

• We'll explain ① ② before wrapping up this chapter.

Thm1 (Archimedean property of $\mathbb{R}$)

$\mathbb{N}$ is unbounded from above in $\mathbb{R}$.

Rmk (1) This is not nonsense! $\mathbb{R}$ is completely artificial. Who knows, maybe some magic number in $\mathbb{R}$ is insanely large.

(2) This result is super useful!

Ex. $S = \{1 - \frac{1}{n} : n \in \mathbb{N}\}$. $\inf S = \min S = 0$. max DNE.

$\sup S = 1$. Why? As $n \to \infty$, $\frac{1}{n} \to 0$, $1 - \frac{1}{n} \to 1$. (*)

Note: the step "$n \to \infty$" depends on Archimedean!

⚠ The part (*) in the above is only a sketch, not a rigorous proof. To make it rigorous, you can try to show:

(1). 1 is an upper bound of S.

(2). Any number $a<1$ cannot be an upper bound (Since $a<1$, $a=1-\varepsilon$ for some $\varepsilon>0$. Choose n large enough s.t. $n>\frac{1}{\varepsilon}$. Then, $\varepsilon>\frac{1}{n} \Rightarrow 1-\varepsilon<1-\frac{1}{n}$.).

Pf. (of Archimedean)

Suppose $\mathbb{N}$ is bounded from above in $\mathbb{R}$. then by CA. $\exists\ m = \sup \mathbb{N}$ in $\mathbb{R}$. This also implies that $m-1$ is NOT an upper bound of $\mathbb{N}$. i.e. $\exists\ n \in \mathbb{N}$ s.t. $m-1<n$. This implies $m<n+1$. Since $n+1 \in \mathbb{N}$. We get a contradiction. □

Prop1. The following statements are equivalent to Archimedean:

① $\forall\ x>0$ and $y \in \mathbb{R}$. $\exists\ n \in \mathbb{N}$ s.t. $nx>y$.

② $\forall\ x>0$. $\exists\ n \in \mathbb{N}$ s.t. $0<\frac{1}{n}<x$.

Pf. (Plan: Archimedean $\Rightarrow$ ① $\Rightarrow$ ② $\overset{HW}{\Rightarrow}$ Archimedean)

(1) Archimedean $\Rightarrow$ ①:

Consider the real numbe $\frac{y}{x}$. By Archimedean, $\exists\ n \in \mathbb{N}$. s.t. $n>\frac{y}{x}$. Hence. $nx>y$.

(2). ① $\Rightarrow$ ②: Take $y=1$ in assumption ①. then we know that $\exists\ n \in \mathbb{N}$ s.t. $nx>1 \Rightarrow x>\frac{1}{n}>0$. □

Prop2. There exists a positive real number x s.t. $x^2=2$.

Pf Define $S=\{r \in \mathbb{R}:\ r>0$ and $r^2<2\}$.

(To be clear. "2" is defined as "1+1". Similarly. we can

define $\mathbb{N}$ as $\{1, 1+1, 1+1+1, \cdots\}$. Everything is rigorous.)

Obs. $S \neq \phi$ and S is bounded from above.

(Obviously. $1 \in S$. And one can show that 2 is an upper bound of S. Indeed. $\forall r \in S$. if $r > 2$. then $r^2 > 2^2 = 4 > 2$. contradiction.)

By CA. $x := \sup S \in \mathbb{R}$ exists. Obviously, $x > 0$. WTS. $x^2 = 2$.

(If $x^2 < 2$. use Prop 1. to show that $\exists n \in \mathbb{N}$ s.t. $x + \frac{1}{n} \in S$. Similarly. if $x^2 > 2$. show that $\exists M \in \mathbb{N}$ s.t. $x - \frac{1}{M}$ is also an upper bound of S. contradicting $x = \sup S$.) □

Rmk. To make sure that eqn "$x^2 = 2$" uniquely defines a positive real number. ("$\sqrt{2}$"). we still need to show that the positive solution to $x^2 = 2$ is unique.

(Suppose $\exists x_1 \neq x_2 \in \mathbb{R}$ s.t. $x_1^2 = 2 = x_2^2$. then by Order axiom. either $x_1 < x_2$ or $x_1 > x_2$.

WLOG. assume $x_1 < x_2$. Then $x_1^2 < x_2^2$. Contradiction.)

- By doing the same thing to any number $a > 0$. (replacing 2 by a). we can define the operation of taking square root.

Thm. (A). $\mathbb{Q}$ is dense in $\mathbb{R}$. More precisely. $\forall x \in \mathbb{R}$. $\forall \varepsilon > 0$.

$\exists r \in \mathbb{Q}$ s.t. $|x - r| < \varepsilon$.

(B). $\mathbb{R} \setminus \mathbb{Q}$ is dense in $\mathbb{R}$.

Rmk. An equivalent way to state density is: $\forall x, y \in \mathbb{R}$ s.t. $x < y$. $\exists r \in \mathbb{Q}$ s.t. $x < r < y$. (prove the equivalence!)

Ex. Use this to show that for $S = \{r \in \mathbb{Q} : 0 < r < \sqrt{2}\}$.

$\sup S = \sqrt{2}$.